

CONNECTING THE ENTIRE ENTERPRISE

Unified, Intelligent, & Secure SD-WAN



TOPICS

What is wide area networking?

How wan has been evolving over years

Let's design typical enterprise network

Issue in traditional enterprises from wan perspective (why sometimes new??)

How these issues can be resolved? SD-WAN introduction

SD-WAN solutions is a big puzzle. Let's break it down, control plane and data plane. Can we separate them? Any benefits? Concept of next

Let's combine all the process together for a complete solution

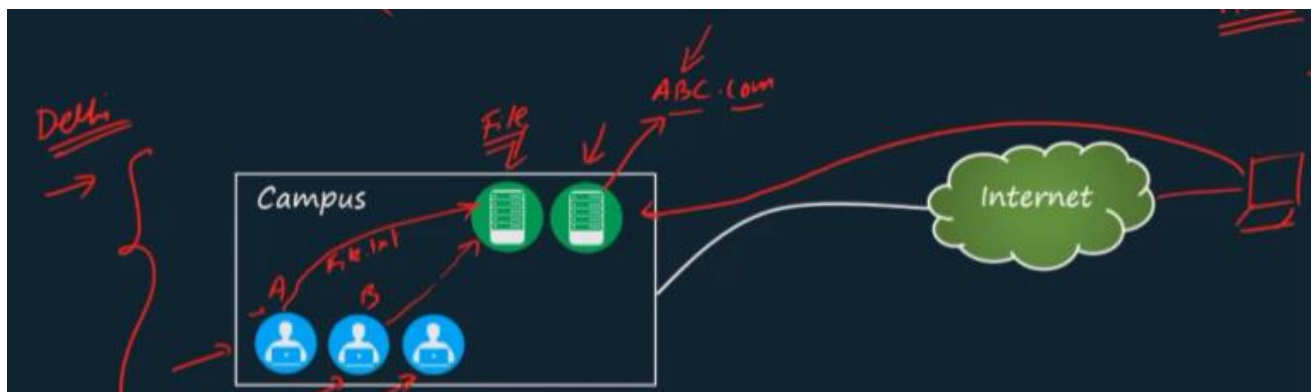
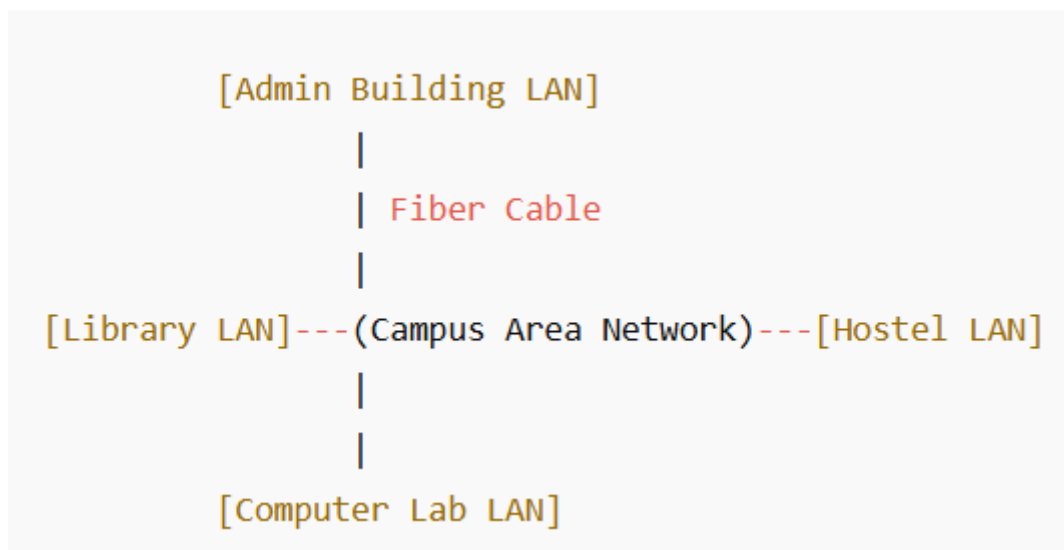
1-CAMPUS AREA NETWORK

What is CAN (Campus Area Network)?

A Campus Area Network (CAN) connects multiple Local Area Networks (LANs) within a limited area like a college campus, business park, or military base. It links buildings that are close to each other (usually within a few kilometers).

Example:

A university network that connects the library, labs, and admin office together — all sharing printers, internet, and resources.



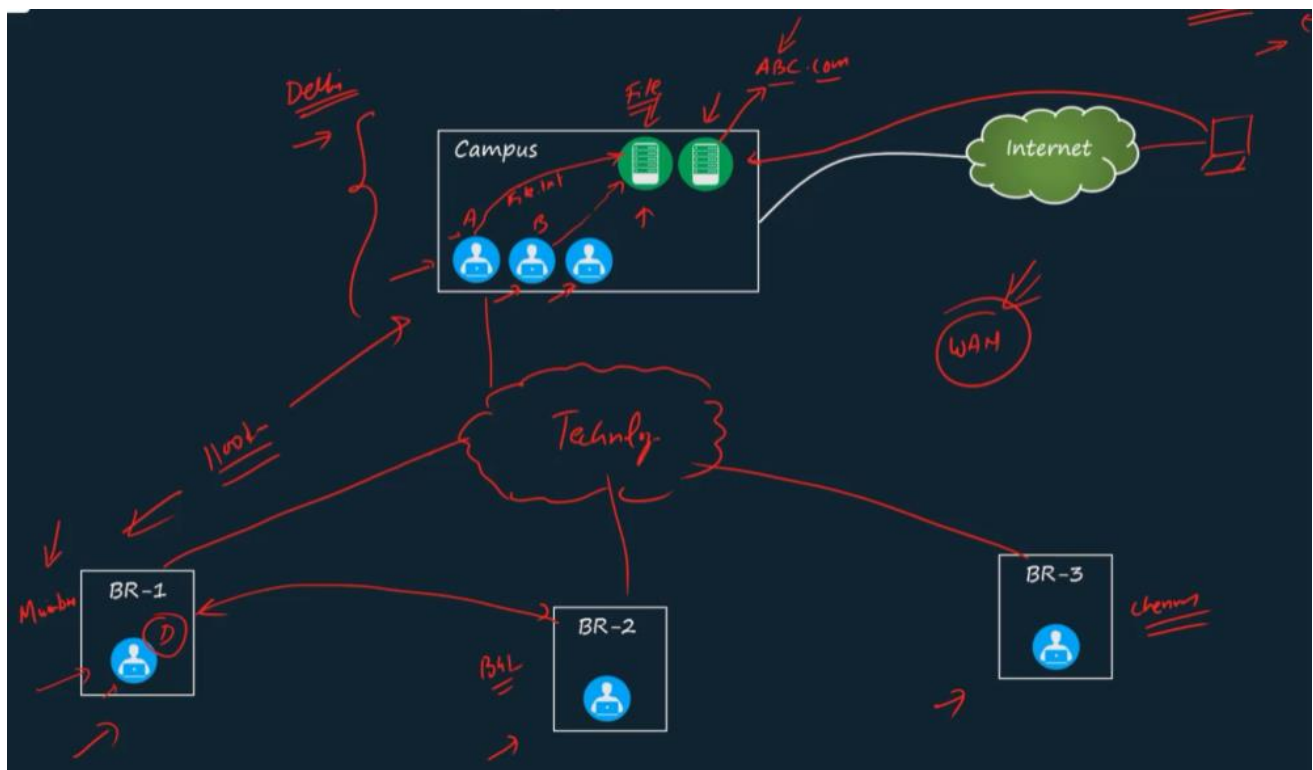
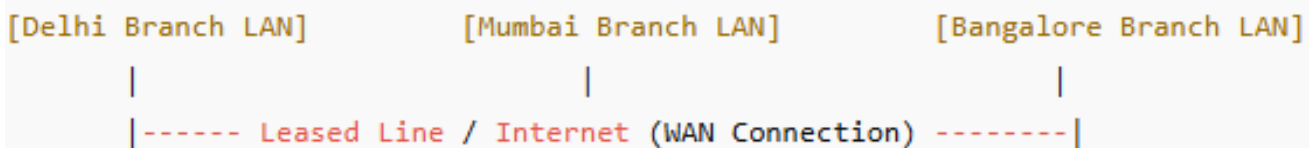
2-WIDE AREA NETWORK

What is WAN (Wide Area Network)?

A Wide Area Network (WAN) connects networks over large distances, like between different cities or countries. It uses public networks (like the internet) or private leased lines.

Example:

A bank network connecting its branches in Delhi, Mumbai, and Bangalore using leased lines or VPN over the internet.



WAN-EVOLUTION

1. Leased lines
2. ATMs and FR
3. MPLS
4. Internet
5. GRE
6. IPSEC DMVPN

1. LEASED LINES:

A leased line is a dedicated, private connection between two locations that is always active and not shared with anyone else. It's used by businesses to connect offices in different places with fast and reliable internet or data transfer.

How It Works:

- The telecom company (like Airtel, BSNL, or Jio) provides a direct cable connection between two sites.
- Unlike normal internet, the bandwidth is fixed and guaranteed (e.g., 100 Mbps always).
- It's secure because no other user shares that line.

[Delhi Office LAN]

|

Leased Line (Private Connection)

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[Mumbai Office LAN]

The routers of both branches are connected through an ISP (has their routers in between this two branches), but they function as if they are directly connected to each other.

3. MPLS:

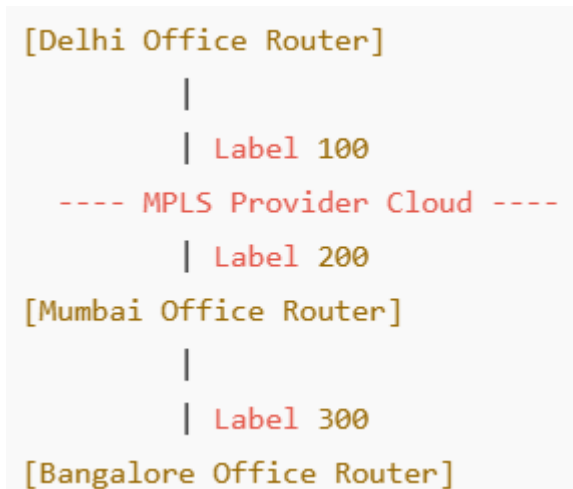
What is MPLS

MPLS is a high-speed, reliable WAN technology that sends data using labels instead of IP addresses. It helps route packets faster and more efficiently through a service provider's network — like express lanes on a highway for your data.

How It Works:

When data enters the MPLS network, it gets a label (a short number).

Routers in the provider's network use this label to forward data quickly — they don't need to check the IP address every time. This creates label-switched paths (LSPs) — fixed routes that ensure speed and reliability.



A company has offices in Delhi, Mumbai, and Bangalore. Each office connects to the service provider's MPLS cloud. When data travels from Delhi to Mumbai, it follows a label-switched path inside the MPLS network — providing QoS (Quality of Service) for voice, video, or critical applications.

4. INTERNET: 18:02

What is INTERNET

When branches use the **internet without SD-WAN or VLAN**, it means each branch uses the **normal public internet** connection — just like your home Wi-Fi — to access **online services**, but **not to connect directly to each other's local networks**.

How It Works:

→ Each Branch Has Its Own ISP Connection

Every branch (like Delhi, Mumbai, Chennai) buys an internet connection from a local ISP (Internet Service Provider)

Example: Airtel in Delhi, Jio in Mumbai.

→ Access Cloud Services or Websites

Employees in each branch use the internet to access email, cloud storage (like Google Drive), CRM systems, or company websites.

These services are hosted on the internet, so every branch can use them without needing a private link.

→ Communication Through Online Platforms

Branch employees communicate using Zoom, Google Meet, or Teams, which all work over the public internet.

→ No Direct LAN-to-LAN Connection

Branches cannot access each other's local files or printers because there's **no VPN or VLAN connecting them privately**.

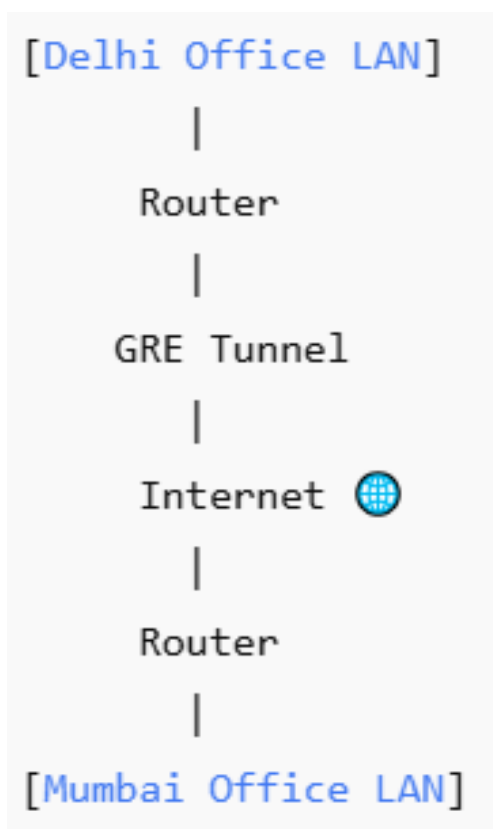
5. GRE (traffic forwarded based routing table)

What is INTERNET

GRE is a tunneling protocol that allows you to encapsulate one network protocol inside another. It is commonly used to connect two networks over the internet or WAN, making them appear as if they are directly connected, even if the connection goes over a public network.

How It Works:

GRE creates a “tunnel” between two routers. Packets from one network are wrapped (encapsulated) in GRE and sent through the tunnel. At the other end, the packets are unwrapped and delivered to the destination network. Works with any Layer 3 protocol (not just IP).



A company has offices in Delhi and Mumbai, and both offices are connected to the internet. A GRE tunnel is established between the routers at the Delhi and Mumbai offices. This tunnel makes it appear as if the two offices are directly connected, allowing employees to access internal servers and resources securely over the tunnel, even though the data travels through the public internet.

5. IPSEC (ACL controls the tunnel traffic)

What is INTERNET

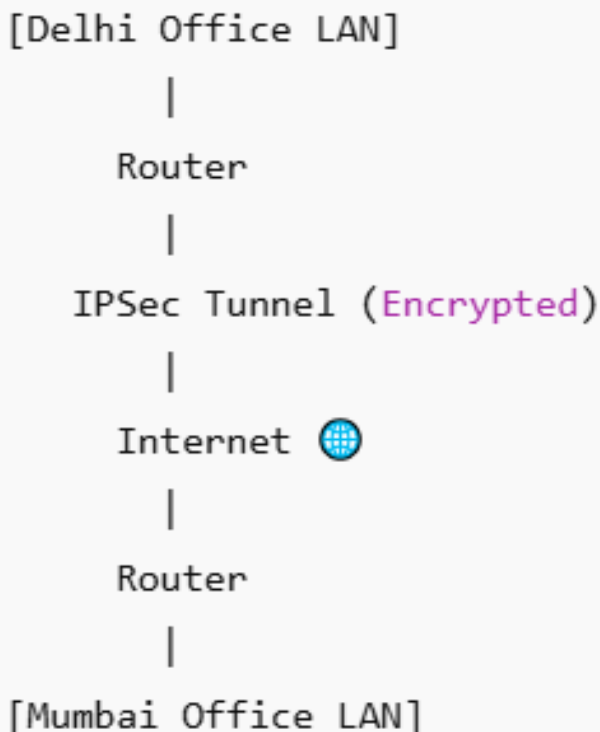
IPSec is a security protocol used to encrypt and protect data sent over a network, especially over the internet or WAN. It ensures that data is confidential, authentic, and tamper-proof while traveling between two locations.

How It Works:

IPSec creates a secure tunnel between two devices (usually routers or firewalls). All traffic passing through the tunnel is encrypted, so outsiders cannot read it.

It provides:

- Encryption – keeps data private.
- Authentication – ensures data comes from the correct source.
- Integrity – ensures data isn't changed in transit.

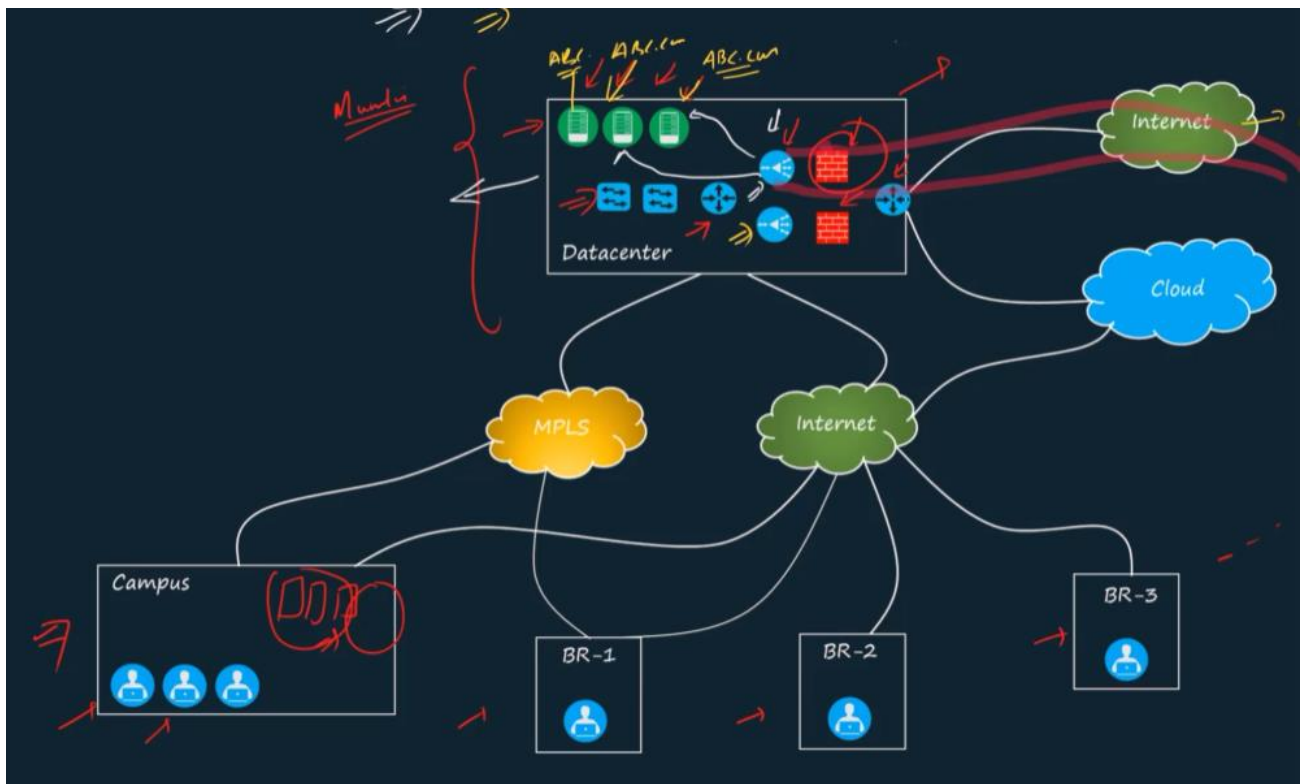


A company has offices in Delhi and Mumbai, and both offices are connected to the internet. An IPsec VPN tunnel is established between the routers at the Delhi and Mumbai offices. All data traveling through this tunnel is encrypted, allowing employees to securely access internal servers and resources over the public internet.

3-Design Of An Enterprise network

What is Datacenter?

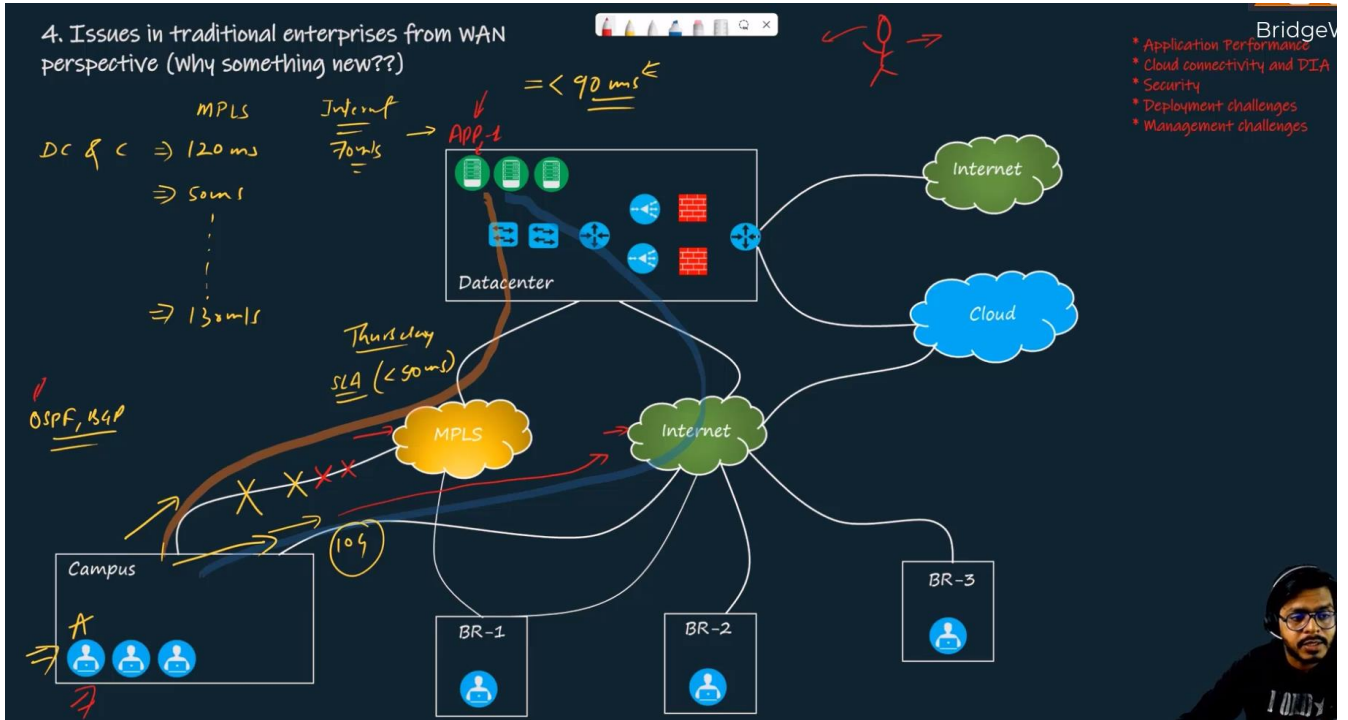
A Data Center is a special facility where a large number of servers, storage systems, and networking equipment are kept together to store, manage, and process data for organizations or online services.



1000-typical enterprise design

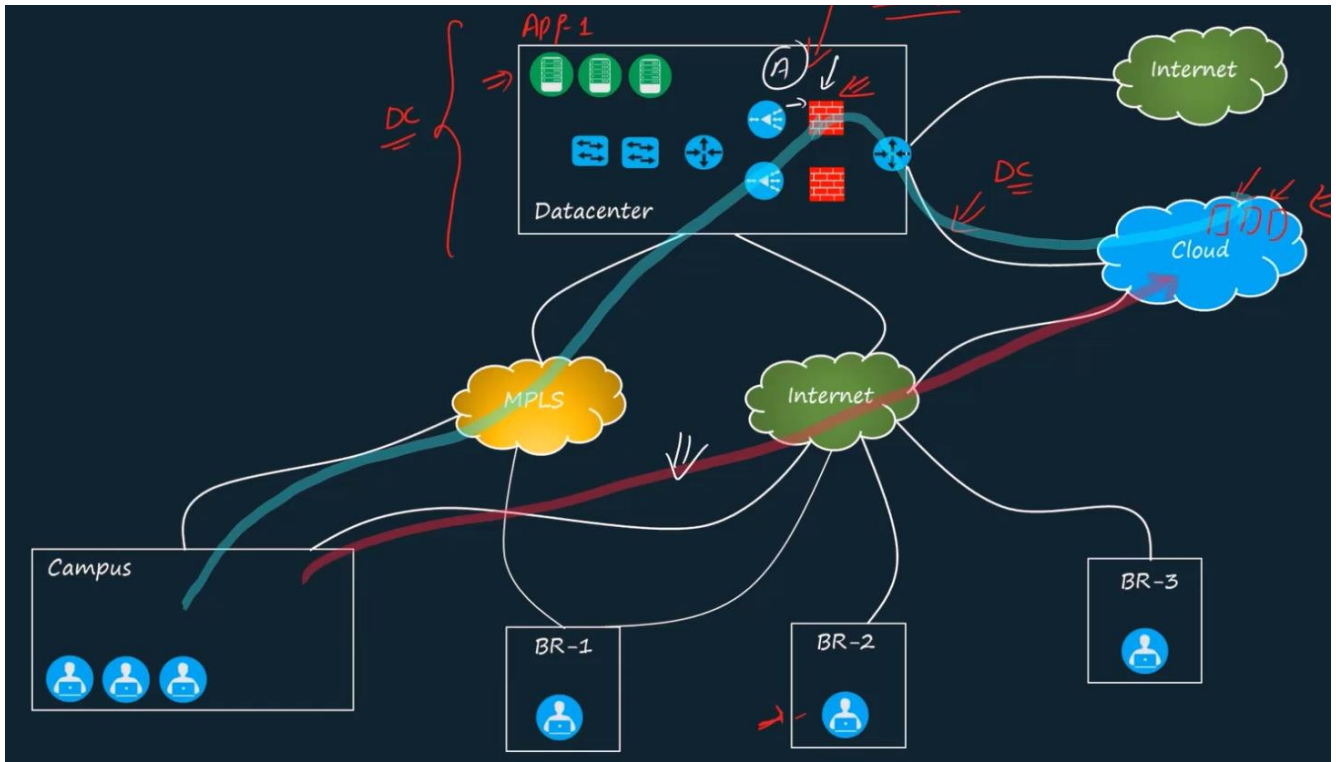
3tier (Acc, Dist, Multi), VXLAN, ACI

Issues Of Traditional Enterprise network (WAN)



IWAN, Performance routing

Issues Of Traditional Enterprise network (WAN)

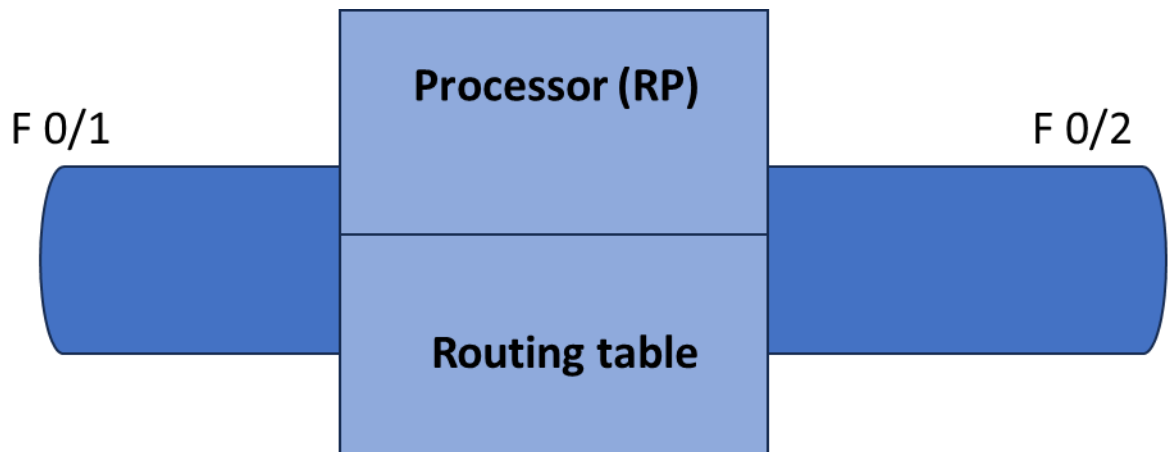
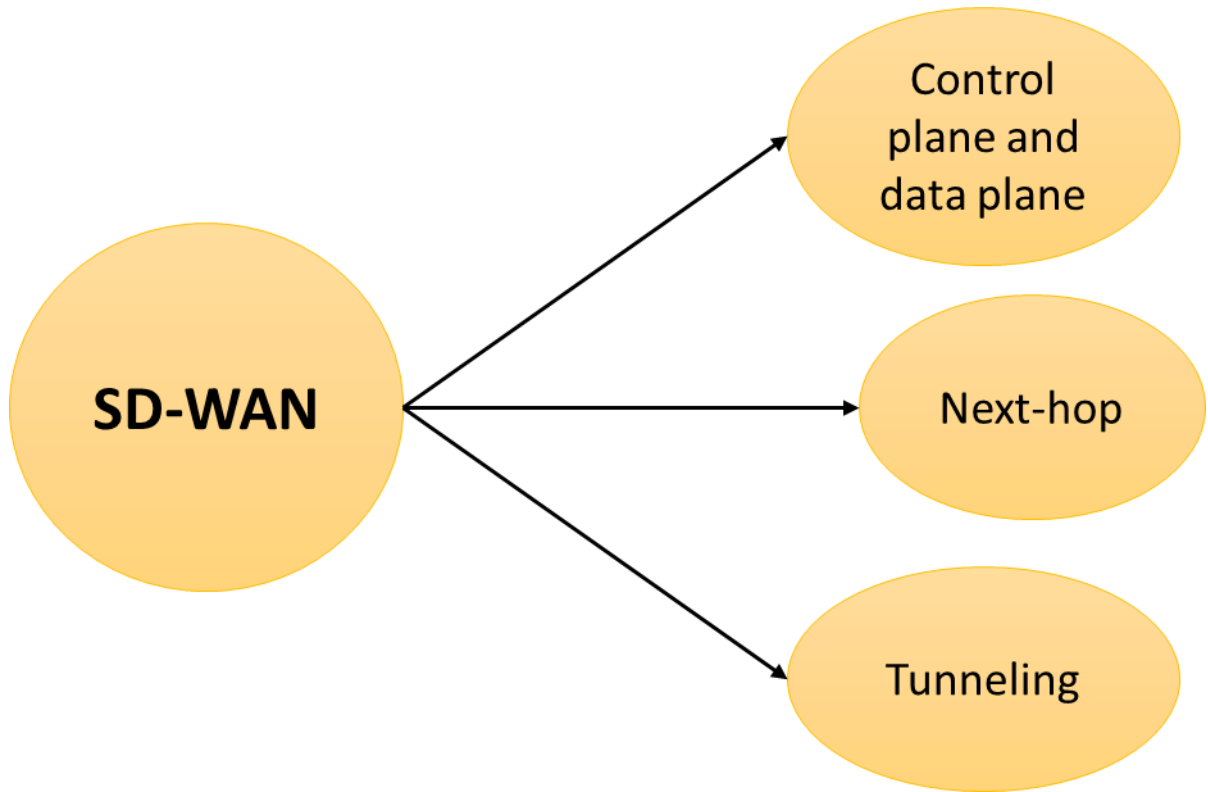


BR1 and 2 data center, but br3 needs a vpn to go to data center

SD-WAN INTRO

1. Delay (Latency)
2. Jitter
3. Packet Loss

We need to feed something on smart devices and need someone to observe on this.



SDWAN – D1 (INTRO – V V V)

TOPICS:

1. SD-WAN solution is a big puzzle:

- Control plane and Data plane. Can we separate them? Any Benefit?
- concept of next-hop of a route
- Tunneling

2. Let's combine all the pieces together for a complete solution

3. Looking slightly deeper into SD-WAN solution. Let's introduce terms

now.

4. The orchestrator: VBOND

5. Let's talk about: Vmanage

1. SD-WAN solution a big puzzle

- i. **CEF** – Cisco Express Forwarding, FIB, LFIB
- ii. **Source Controller**, other network devices (Router, Switch)

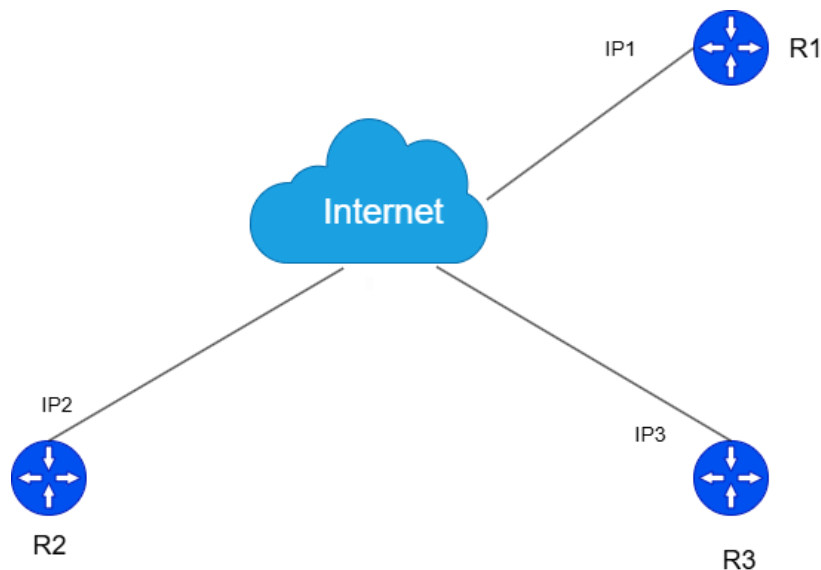
Source Controller – handles all control information of those routers.

Other devices – handles all data plane by themselves after getting information from the Source controller

How it works:

Each of the routers sends their routing table. Source device process the information and send appropriate or whole routing table to routers.

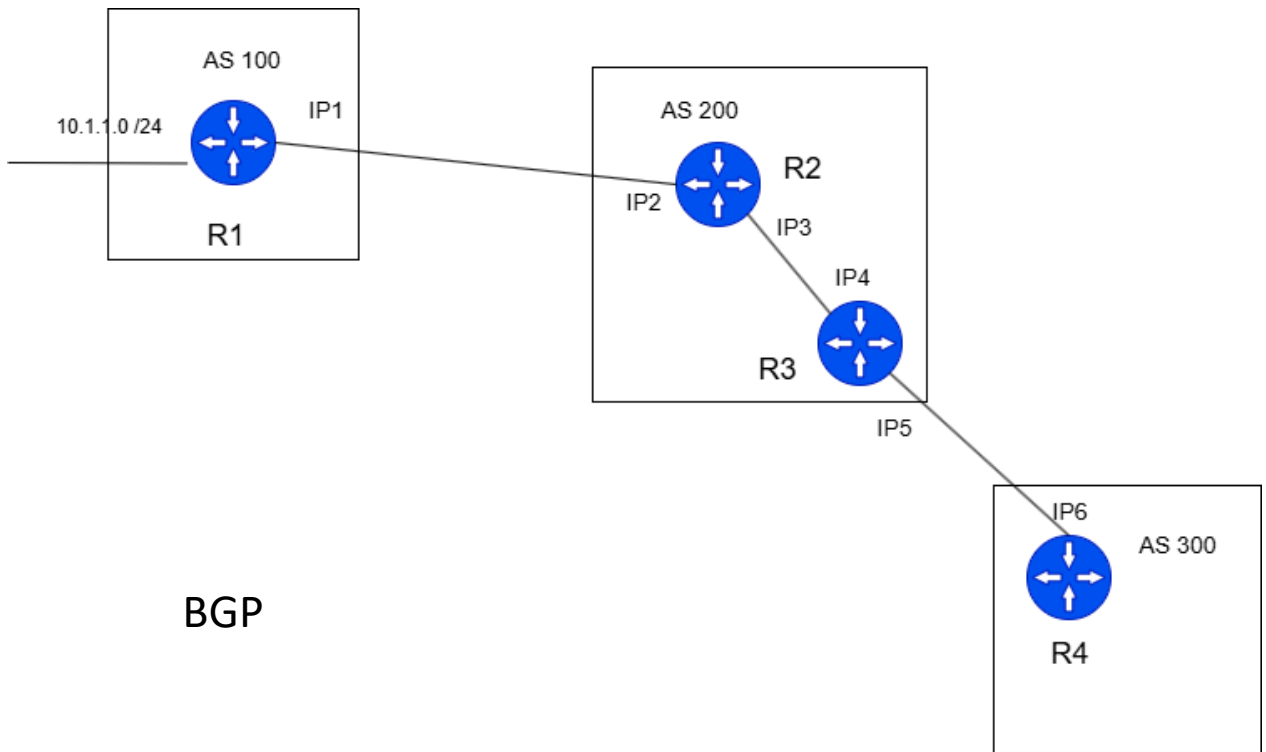
2. Concept of Next-Hop



I want that R2 should always reach R2 via R1.

Full Mesh – already running – 1.1.1.0/24 via IP1

Hub & Spoke – Update need – 1.1.1.0/24 via IP3



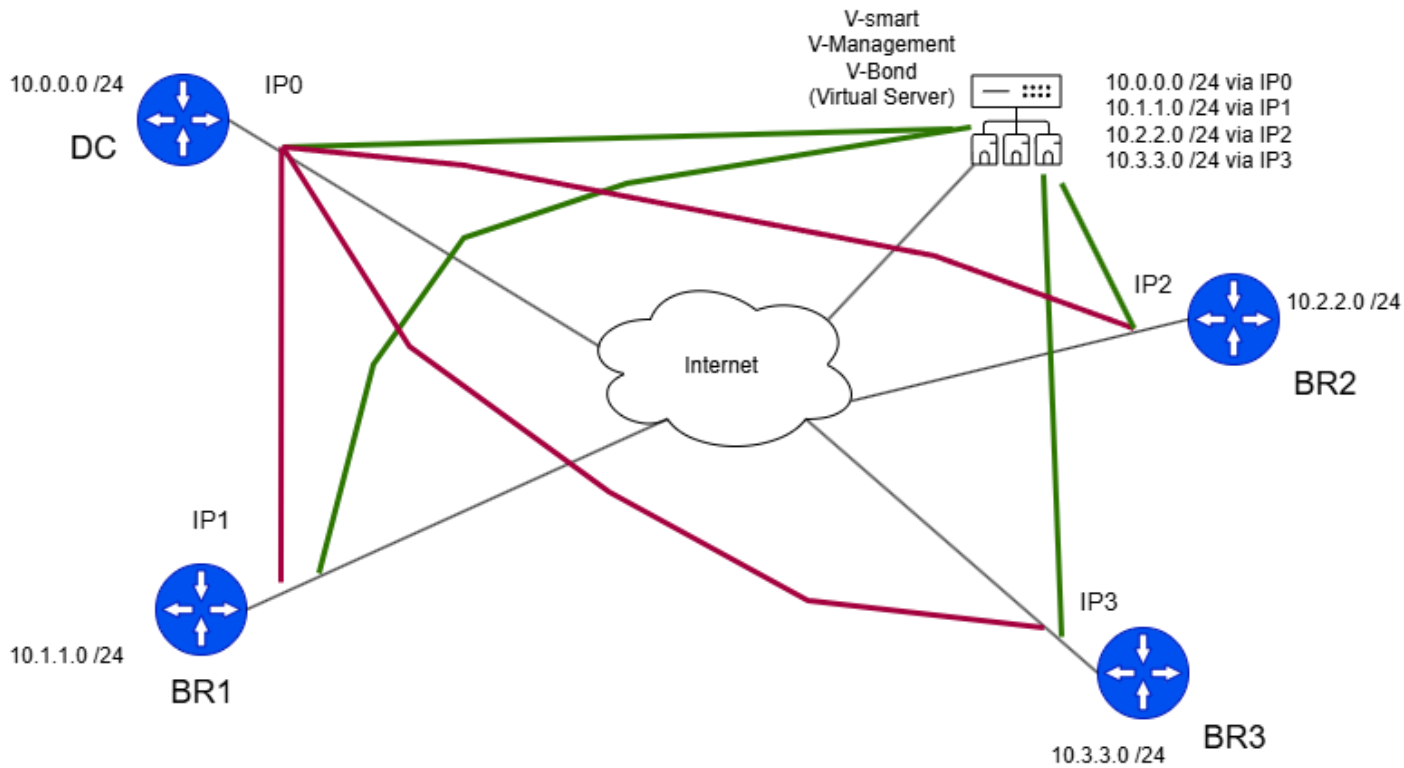
BGP

What is the next-hop for 10.1.1.0 network on every router?
R2 – IP1, R3 – IP1, R4 – IP1

BGP isn't about sharing routes between two router, it is about sharing routes between 2As.

3. Tunneling (gre) – Already done

4. Let's combine all the above



When the routers connected to the vSmart become active, they establish secure control tunnels with the vSmart controller. The **vSmart** then processes the received control connections and propagates the routes, setting the data center as the next-hop.

Vmanagement – It is application which is used to control the vsmart like configuration, monitoring, and troubleshooting. Configurations are pushed from Vmanagement to vsmart.

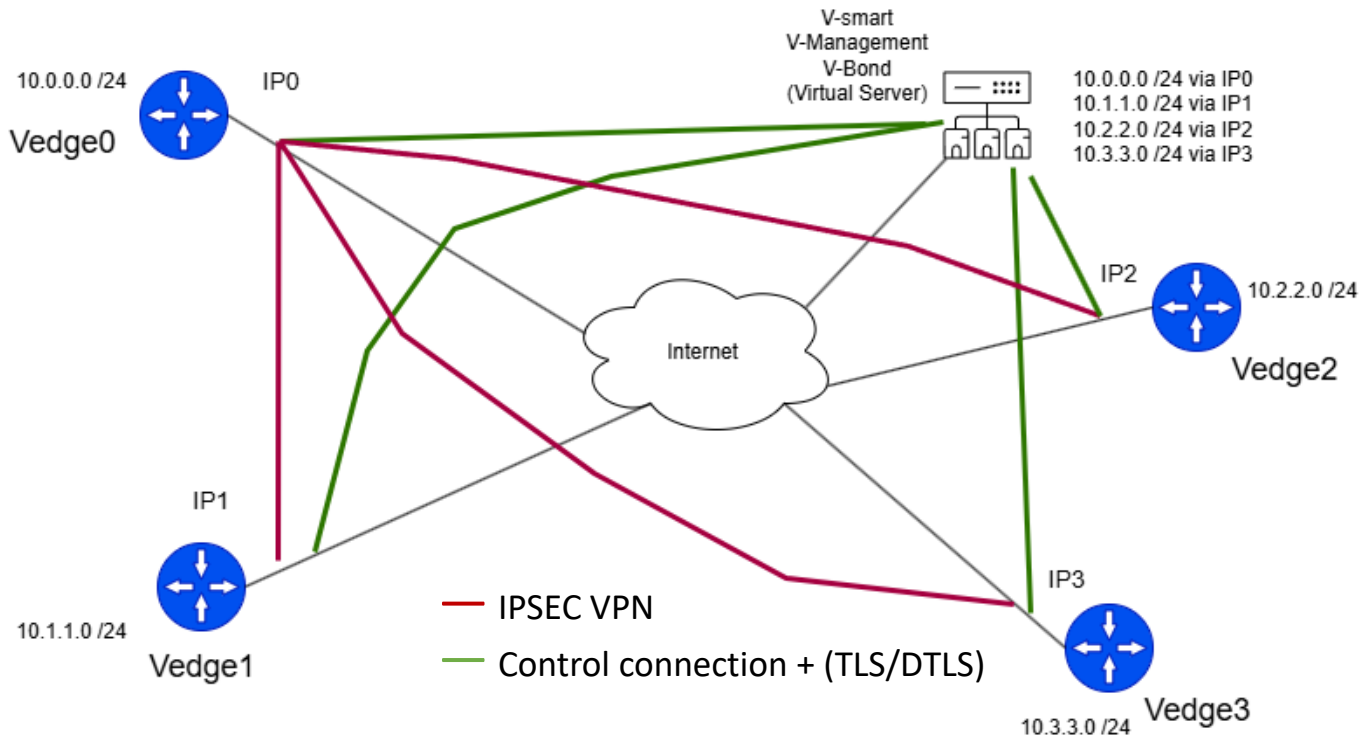
Vbond – It is responsible to give the ip address of Vsmart and Vmanagement to the **Branch and DataCenter**.

SD-WAN (D-2)

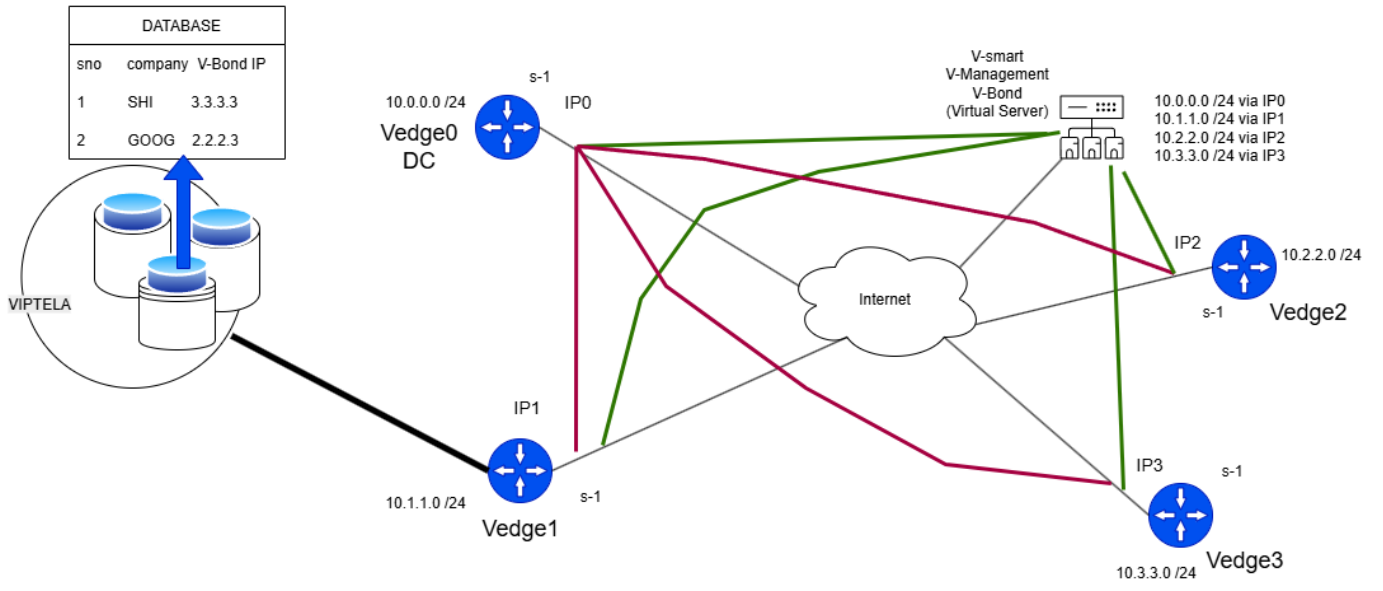
TOPICS:

1. The orchestrator: Vbond
2. Let's talk about Vmanage
3. What we see in cisco's documentation
4. Why we are learning network security in SD-WAN?
5. Basics of network security or CIA

1. The orchestration: VBond



VBonds of VBonds



Orchestration: Someone who can arrange things or combine things to get desired result.

Zero-Touch solution, plug and play -- when we power on this device, it create tunnel and connection everything by itself.

How a device get configuration after attaching to SD-WAN?

1. Device Boot & Initial Setup

When a new device is powered on:

You manually configure only a few bootstrap parameters, like:

system-ip 10.10.10.10

site-id 100

organization-name MyOrg

vbond <vbond-ip>

This minimal config allows the device to find and authenticate with the SD-WAN controllers.

2. vBond Orchestration – Authentication & Control Setup

The vBond orchestrator is the first controller the device contacts.

It helps the device discover and authenticate with:

vManage (management plane)

vSmart (control plane)

vBond ensures all devices trust each other using certificates issued by the SD-WAN PKI.

Result:

Device now forms secure DTLS/TLS control connections to vSmart and vManage.

3. vManage Delivers Configuration (via NETCONF)

Once authenticated:

vManage detects the new device (by its UUID and serial number) in its inventory (pre-added via the Smart Account or PnP portal).

Admin associates a Device Template and Feature Templates to that router.

vManage generates XML configuration from those templates. It then pushes this config to the router using NETCONF over TLS.

Protocol used:

NETCONF (TCP port 830) → secure configuration channel.

Configuration includes:

Interface settings (VPN 0, VPN 512, etc.)

Routing (OMP, BGP, static)

Policies (control/data plane)

System parameters

4. vSmart Delivers Routing & Policy (via OMP)

After configuration is applied:

The router forms an OMP (Overlay Management Protocol) session with vSmart.

vSmart then sends routes and policies (centralized control policies, TLOCs, etc.) to the router.

Protocol used:

OMP over DTLS/TLS (control plane communication)

5. Operational State

Now the device:

Has its full configuration (from vManage)

Knows routing and policy information (from vSmart)

Can form secure IPsec data tunnels with other SD-WAN sites.

Data Plane Protocol:

IPsec tunnels built automatically based on OMP TLOC advertisements.

Summary Table

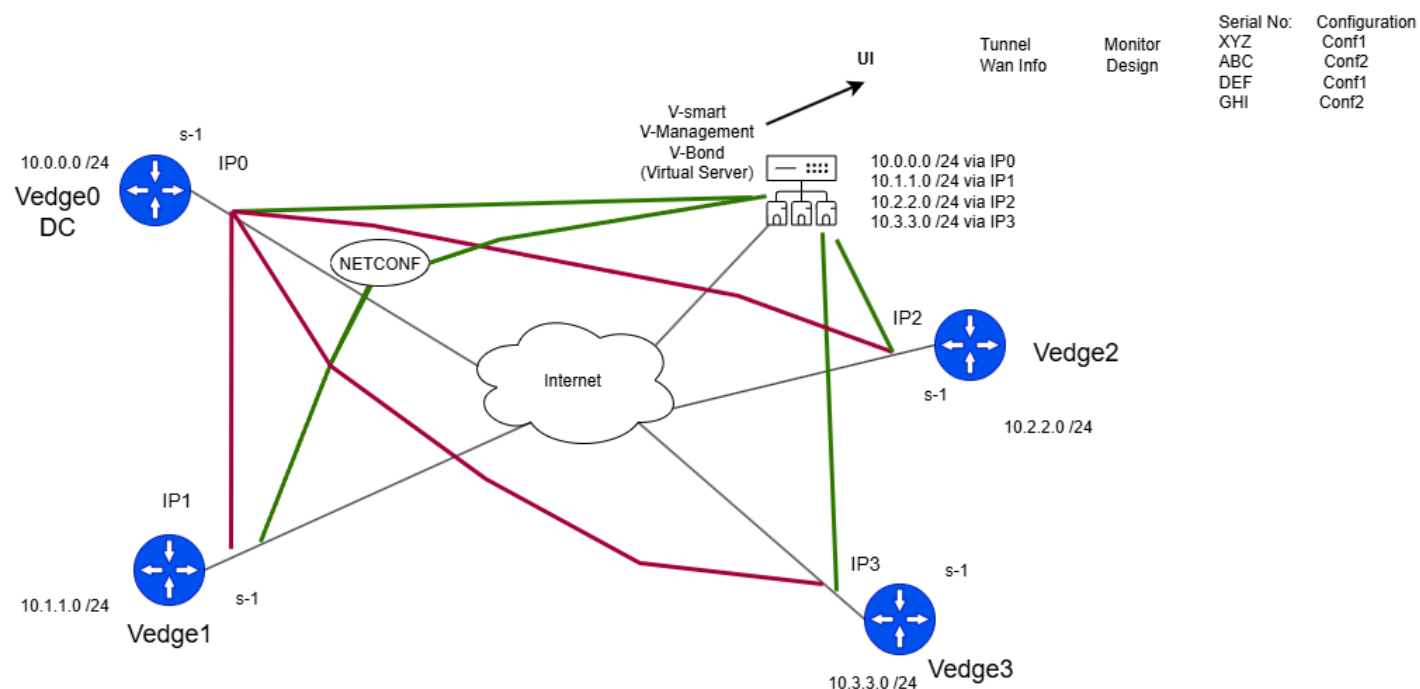
Step	Component	Function	Protocol
1	Device Boot	Bootstrap config	—
2	vBond	Authentication & controller discovery	DTLS/TLS
3	vManage	Push configuration	NETCONF
4	vSmart	Distribute routes/policies	OMP
5	Data Plane	Site-to-site encrypted tunnels	Ipsec

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All together:

Step	Component	Function	Protocol
1	Device Boot	Bootstrap config	—
2	vBond	Authentication & controller discovery	DTLS/TLS
3	vManage	Push configuration	NETCONF
4	vSmart	Distribute routes/policies	OMP
5	Data Plane	Site-to-site encrypted tunnels	IPsec

2. Vmanage: Single pane of Glass



The **vManage** platform is the centralized Network Management System (NMS) for the Cisco SD-WAN (Software-Defined Wide Area Network) solution. It serves as the single pane of glass for all day-to-day operations, including deployment, configuration, monitoring, and troubleshooting of the entire SD-WAN fabric.

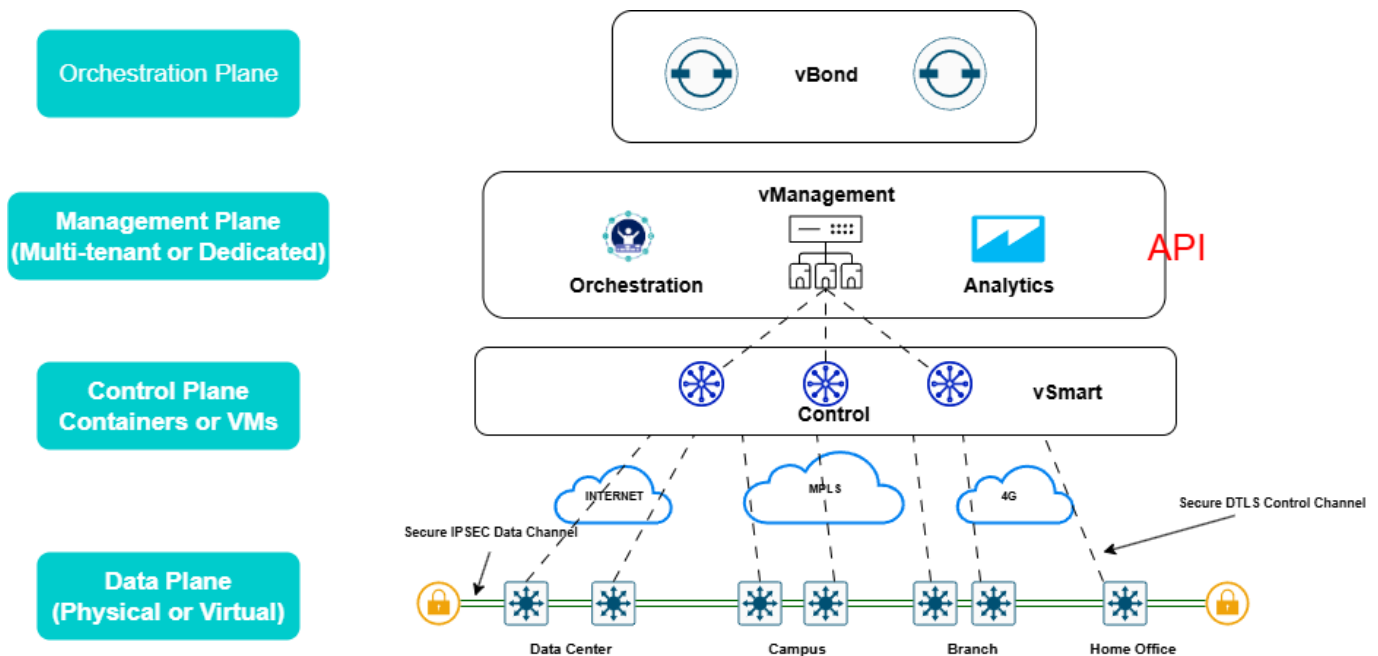
In the SD-WAN architecture, vManage is the component that belongs to the **Management Plane**.

It provides a single graphical user interface (GUI) to configure and manage all devices in the SD-WAN overlay network, including WAN Edge routers, vSmart Controllers, and vBond Orchestrators.

Administrators use vManage to create configuration templates and policies, which are then pushed out to the devices. By applying a template, a device is switched to **vManaged mode**, where its configuration is centrally controlled by vManage, replacing local CLI configuration.

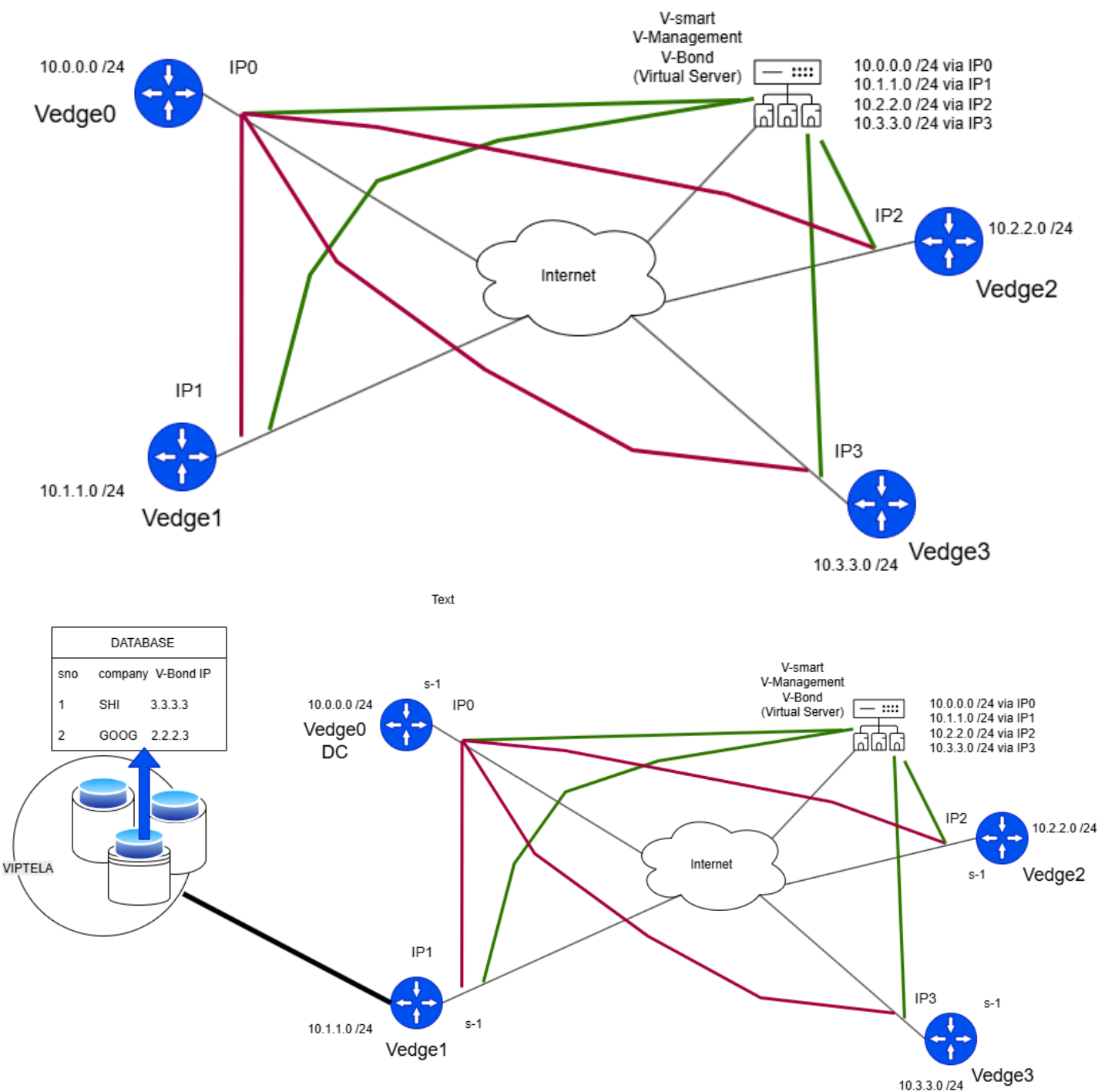
It utilizes protocols like **NETCONF/YANG** models to orchestrate policies to the rest of the SD-WAN infrastructure.

3. What we see in Cisco's Documentation



vEdge and **cEdge** are the two primary types of WAN Edge Routers. These devices sit at the perimeter of a site (like a branch office, data center, or cloud) and perform the data plane functions, forming the secure overlay fabric.

2. Concept of Next-hop



Orchestration: Someone who can arrange things or combine things to get desired result.

Zero-Touch solution, plug and play -- when we power on this device, it create tunnel and connection everything by itself.